

Exploring RGB and Video V-JEPA 2.1 Features for Aerial Depth Estimation and Semantic Scene Completion

Table 6. Evaluation of frozen foundation-model features with increasingly capable monocular depth heads and posed multi-view stereo integration.

Method	$\delta_1 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	SILog $\downarrow$
VJ DPT	0.747	0.162	6.730	0.200
VJ video Linear Probe	0.701	0.191	7.055	0.207
D3 Linear Probe	0.782	0.156	6.097	0.186
D3 FiLM-DPT	0.848	0.121	4.358	0.161
VJ DTA-MVS (pending)	—	—	—	—
VJ DTA-FiLM-MVS	0.768	0.175	6.406	0.211
MVSF++/D2-B	0.748	0.231	7.657	0.195
MVSF++/VJ-G	0.761	0.225	7.517	0.193
MVSF++/D3-H+	0.828	0.136	4.611	0.130

Table 7. Altitude interpolation diagnostics.

Method	$\delta_1 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
A. Unseen altitude; familiar scenes					
Tr: 01–08@30/50 m; Val: 01–04@40 m; Te: 05–08@40 m					
DPT	0.789	0.151	6.728	5.775	0.180
FiLM-DPT	0.965	0.102	4.707	3.957	0.120
B. Unseen scenes and altitude					
Tr: 01–04@30/50 m; Val: 05–06@40 m; Te: 07–08@40 m					
DPT	0.925	0.102	4.807	3.994	0.125
FiLM-DPT	0.707	0.205	8.506	7.918	0.169

Table 8. FiLM conditioning ablation. Test mean $\pm$ SD over seeds 0/1/2. Numerically best mean in bold.

Conditioning	$\delta_1 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	SILog $\downarrow$
DPT	.728 $\pm$.034	.174 $\pm$.006	6.709 $\pm$.385	.202 $\pm$.010
Altitude FiLM	.798 $\pm$.020	.145 $\pm$.005	5.236 $\pm$.322	.180 $\pm$.006
Intrinsics FiLM	.711 $\pm$.015	.182 $\pm$.004	7.115 $\pm$.243	.216 $\pm$.007
Alt.+intr. FiLM	.810$\pm$.006	.137$\pm$.005	4.907$\pm$.140	.171$\pm$.005

Table 9. Extension of main-paper Table 5 with MVSFormer++ depth into FSSC geometric layer.

Variant	SC IoU (%) $\uparrow$	SSC mIoU (%) $\uparrow$
DINOv3 context + DINOv3 MVS depth	43.37	4.47
DINOv3 context + DINOv3 MVS depth + VoxDet V5-V2	48.06	4.84
V-JEPA2.1 context + V-JEPA2.1 MVS depth	35.02	3.28

We thank all reviewers for their constructive feedback. VJ/D2/D3/DA2/MVSF abbreviate V-JEPA 2.1/DINOv2/DINOv3/Depth Anything 2/MVSFormer++; response table numbering starts at 6, continuing submission.

Responses common to LD5V, CWVJ, and y7qY: Table 6 caption is appropriate for the Table 3 and it extends Table 3 by reporting corrected result for DPT, Spatial alignment included video methods, suggested DINO variants for faithful comparison and attribution of merits or flaws, requested MVS(simple) and supportive VJ/D features backed MVSF++(sophisticated) variants depth results. Only important baseline is DA2-Occufly. For other baselines refer table 5 of occufly paper. We omit superiority claim of VJ FiLM-DPT. Considering the Table 3 and Table 6 we show increasingly capable depth heads and MVS integrations. Although DAv2-OccuFly achieves the strongest reported SiLog and threshold accuracies, foundation models with general pretraining objectives remain competitive and outperform it on absolute-error metrics. Any remaining gap may reflect DAv2’s depth-specific pretraining and end-to-end OccuFly adaptation, whereas frozen VJ/D backbones are adapted only through their depth heads.

Table 7 shall be Table 4. The main purpose of this non standard split ablation is to specifically study methods ability to interpolate to unseen altitude cues. But as rightly pointed there is inherently visual cues leakage. To address that we will report a ablation of unseen scenes and altitude for comparison. A clear architecture diagram for SSC concept, prioritizing data-flow and various components involved can replace Figures 1 and 5 in the final submission.

LD5V:statistical analysis of robustness, and evidence for the central claims. contribution of the FiLM ablation: We provide a matched FiLM ablation and its per-seed results in Table 8 and Table ???. spatial misalignment: We thank the reviewer for pointing this out and adopted 2D feature alignment like frozen RAFT flow to align neighboring features to the target frame; Alongside simple MVS baselines, extending with MVSFormer++, a sophisticated architecture combining cross-view fusion with attention-based feature encoding and cost-volume regularization for depth estimation, is appropriate. Ablations replacing the original D2 backbone with VJ and D3 for depth and FSSC-style geometric lifting are reported in Table 9. Replacing the control’s FoundationSSC decoder/objective with the corrected VoxDet V5-V2 decoder, full-resolution class-balanced VoxNT, and long-tail loss reaches 48.06 SC IoU and 4.84 union-present SSC mIoU; its GT-present SSC mIoU is 5.11. The native MVS posterior remains frozen, so scheduled SSC-to-depth gradients are not active in this treatment. Oracle depth:Replacing estimated depth with

056 GT metric depth in the same geometric lifting and SSC
057 pipeline yields SC IoU/SSC mIoU of 49.40/7.25 with
058 V-JEPA2.1 ViT-G RGB context and 42.01/8.40 with
059 DINOv3 ViT-H/16+ RGB context. per-class results:
060 Already provided in the supplementary material section
061 B. Will be further extended with MVSF++FSSC
062 variants.

063 CWVJ:missing cost figures and related work, existing
064 aerial datasets and multi-dataset evaluation, framing
065 the conclusion as a study rather than a new SSC archi-
066 tecture, the confounded key baseline, and evaluation
067 on only one benchmark. DINOv3: We provide an ab-
068 lation table containing DINOv3 alongside V-JEPA 2.1.
069 This important point also motivates, if permitted, a re-
070 visited title: “Exploring Foundation-Model Spatial and
071 Temporal Features for Aerial Depth Estimation and
072 Semantic Scene Completion.” Single-run:Table ?? for
073 FiLM contribution ablations given the limited rebuttal
074 period, and can extend this analysis to all published re-
075 sults. Because this preliminary study examines frozen
076 foundation features for MDE and SSC rather than on-
077 board deployment, we defer SWaP profiling to future
078 UAV-suitable student models.The mentioned valuable
079 related work will be appropriately cited. existing aerial
080 datasets: The listed datasets are not directly usable
081 for SSC evaluation, where occufly dataset outshines as
082 they provide ground truth voxel grids. Few of the men-
083 tioned datasets especially the ones with depth maps
084 can ofcourse be benchmarked for depth estimation.

085 y7qY: The modest SSC mIoU suggests limitations
086 beyond depth estimation: even the GT-depth or-
087 acle reaches only 7.25/8.40 mIoU, while OccuFly’s
088 96.109The negative video result exposes limitations of
089 our temporal adapter rather than temporal cues them-
090 selves, motivating motion/pose-aligned and occlusion-
091 aware feature fusion used in recent temporal SSC.
092 The MVSFormer++-based FSSC gains are consistent
093 with its cross-view fusion and attention-based cost-
094 volume regularization providing a stronger geometric
095 prior than our simple MVS module. We have consid-
096 ered only scenes 1–8 for non standard altitude inter-
097 polation ablation.